

Engaging with Dictionaries and Sets in Python

In the first part of this series on dictionaries and sets, we explored the usage of dictionaries. In the second article in this series, we take a deep dive into the fascinating world of data structures to gain invaluable behind-the-scenes knowledge and help you broaden your understanding of set and hash.



A set is a collection of unique objects. Set elements need to be hashable. Frozen sets are a part of sets that are immutable.

Creating a set

Here are a few methods to create a set:

- `a = set([1, 2, 3])`
- `b = {1, 2, 3}` #Set literal

A set created using set literals is faster. Calling set constructor requires building a list and finally calling set, whereas set literal directly calls specialised BUILD_SET bytecode. Set doesn't support index operations as it is not built as a list but by using hash.

```
s = {1, 2, 3}
print(s[0])
```

```
Traceback (most recent call last):
  File "dict_set_sample.py", line 44, in <module>
    print(s[0])
TypeError: 'set' object is not subscriptable
```

- **Frozen set**

Here's an example of an immutable version of a set:

```
cars = set(['ford', 'renault', 'tata'])
print(cars)
```

```
cars.add("kia")
print(cars)
```

```
frozen_cars = frozenset(['ford', 'renault', 'tata'])
print(frozen_cars)
```

```
frozen_cars.add("kia")
print(frozen_cars)
```

Output:

```
{'tata', 'renault', 'ford'}
{'tata', 'renault', 'kia', 'ford'}
frozenset({'tata', 'renault', 'ford'})
```

Traceback (most recent call last):

```
File "sample.py", line 121, in <module>
    frozen_cars.add("kia")
AttributeError: 'frozenset' object has no attribute 'add'
```

- **Set comprehensions**

Similar to any list/dict comprehensions, we have a few different types of samples explained in the blog at <https://www.pythonforbeginners.com/basics/set-comprehension-in-python>. Here's an example:

```
curSet = set([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])
newSet = {element*3 for element in curSet}
print("The Current set is:")
print(curSet)
print("The Newly Created set is:")
print(newSet)
```

Output:

```
The Current set is:
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
The Newly Created set is:
{3, 6, 9, 12, 15, 18, 21, 24, 27, 30}
```